

EE 210L: Digital Circuits and Logic Design Lab

Lab 9: Sequential Logic in Verilog

Name: _____

Date: _____

Partner(s): _____

Section: _____

Keep these designs. You will put them on the FPGA board in Lab 11.

Part 5.1: D Flip-Flop



Figure 1: D flip-flop waveform. Mark one edge where d changed early but q waited.

Does q ever change between clock edges?

Part 5.2: Synchronous vs Asynchronous Reset



Figure 2: Both reset styles, reset asserted mid-cycle.

Reset style	Responded immediately?	Waited for clock edge?	Caught the short pulse?
Asynchronous			
Synchronous			

Which 74LS chip from Lab 6 or 7 matches each style?

When would missing a short reset pulse be a bug, and when would it be a feature?

Part 5.3: Blocking vs Non-Blocking

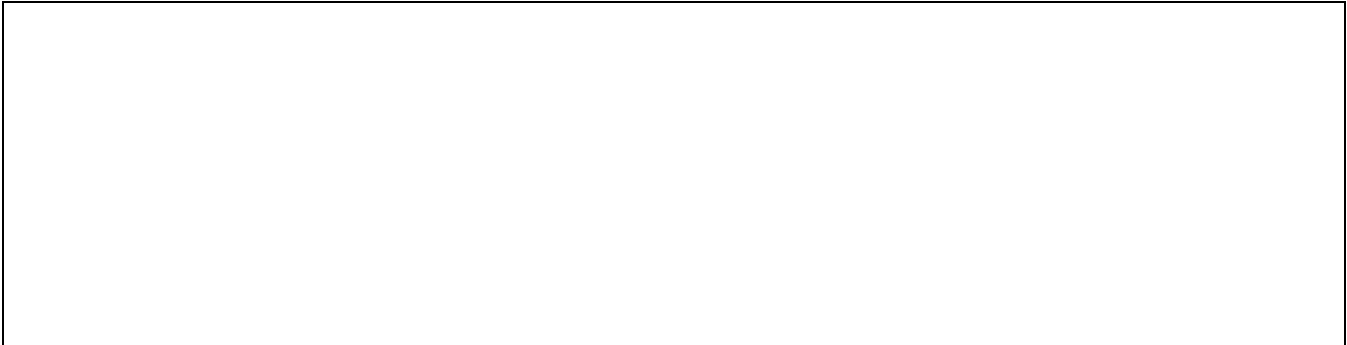


Figure 3: Both shift registers on one waveform.

Clock edge	Non-blocking (<=)		Blocking (=)	
	q1	q2	q1	q2
start				
1				
2				
3				

Version	Flip-flops in the synthesized schematic
Non-blocking	
Blocking	

Did the results match your pre-lab prediction? Explain the hardware difference in your own words.

Part 5.4: Four-Bit Counter

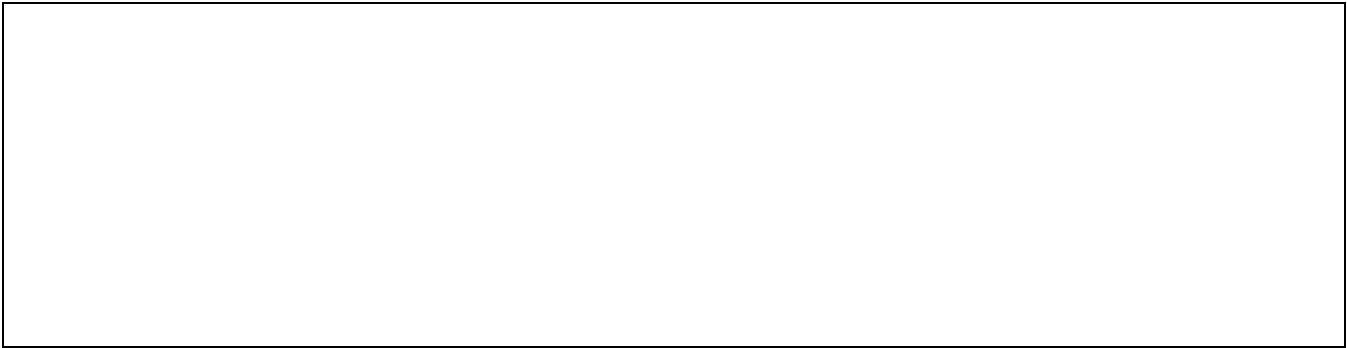


Figure 4: Counter wrapping from 1111 to 0000.

Self-checking result:

Does en = 0 hold the count?

Part 5.5: Shift Register with Modes

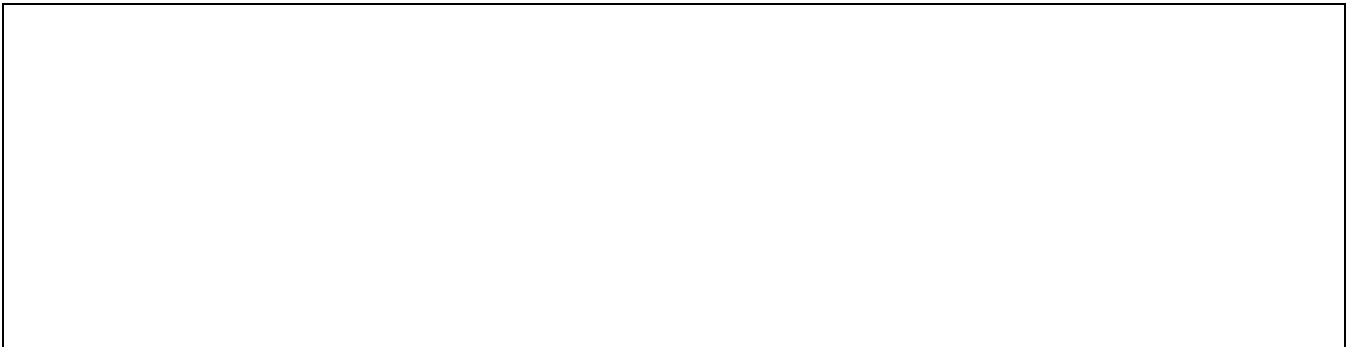


Figure 5: All four modes exercised in sequence.

Mode	Expected behavior	Observed behavior	Match?
00 hold			
01 shift right			
10 shift left			
11 load			

Does this match what you measured on the 74LS194 in Lab 7?

Part 5.6: Ring Counter



Figure 6: Ring counter, at least eight clock edges.

Period of the pattern: _____ clock edges

Discussion and Conclusions

Address at least: the rule for when to use $\leq$ versus $=$, why non-blocking matches what real flip-flops do, and which reset style you would choose inside an FPGA.
